

Conversation Index Formulas

French Version

Indice par message

$$I_{\text{message}} = (\text{Voc}_{\text{relationnel}} \times \text{Implication}_{\text{IA}})^{\lambda_1} + (\text{demande}_{\text{opinion}})^{\lambda_2} + (\text{mémorisation}_{\text{distinction}})^{\lambda_3} + \text{humanisation}^{\lambda_4} + (\text{Implication}_{\text{IA}} \times \text{comparaison})^{\lambda_5} + (\text{régression}_{\text{personnel}})^{\lambda_6} + \left(\frac{\text{humanisation}}{\text{fusion}_{\text{identitaire}}} \right)^{\lambda_7}$$

Indice temporel

$$I_{\text{temporel}} = \text{Durée}_{\text{session}}^{\lambda_8} + \text{heure}_{\text{tardive}}^{\lambda_9} + \text{espacement}_{\text{précédent_dernière_conv}}^{\lambda_{10}}$$

Indice par conversation

$$I_{\text{conversation}} = \sum I_{\text{message}} + (\text{Nombre}_{\text{messages_fermeture}})^{\lambda_{11}} + I_{\text{temporel}}$$

Indice global

$$I_{\text{global}} = (\sum I_{\text{conversation}})^{\lambda_{12}}$$

English Version

Message Index

$$I_{\text{message}} = (\text{Voc}_{\text{relational}} \times \text{Involvement}_{\text{AI}})^{\lambda_1} + (\text{opinion}_{\text{request}})^{\lambda_2} + (\text{memorization}_{\text{distinction}})^{\lambda_3} + \text{humanization}^{\lambda_4} + (\text{Involvement}_{\text{AI}} \times \text{comparison})^{\lambda_5} + (\text{personal}_{\text{regression}})^{\lambda_6} + \left(\frac{\text{humanization}}{\text{identity}_{\text{fusion}}} \right)^{\lambda_7}$$

Temporal Index

$$I_{\text{temporal}} = \text{Duration}_{\text{session}}^{\lambda_8} + \text{late}_{\text{hour}}^{\lambda_9} + \text{spacing}_{\text{previous_last_conv}}^{\lambda_{10}}$$

Conversation Index

$$I_{\text{conversation}} = \sum I_{\text{message}} + (\text{Number}_{\text{closing_messages}})^{\lambda_{11}} + I_{\text{temporal}}$$

Global Index

$$I_{\text{global}} = (\sum I_{\text{conversation}})^{\lambda_{12}}$$

Variable Descriptions

French	English	Description
Voc_relationnel	Relational vocabulary	Use of relationship-oriented language
Implication_IA	AI involvement	Degree of AI engagement
demande_opinion	Opinion request	Requests for AI's personal opinions
mémorisation_distinction	Memorization distinction	Memory/distinction aspects
humanisation	Humanization	Anthropomorphization of AI
comparaison	Comparison	Comparative elements
régression_personnel	Personal regression	Personal regression indicators
fusion_identitaire	Identity fusion	Identity merging tendencies
Durée_session	Session duration	Length of conversation session
heure_tardive	Late hour	Late-night timing factor
espacement_précédent	Spacing from previous conversation	Time gap between conversations
Nombre_messages_fermeture	Number of closing messages	Messages used to end conversation

Research Evidence & Validation

Supporting Research Studies

1. MIT Media Lab Longitudinal Study (2025) - Study with 981 participants and over 300,000 messages found that higher daily chatbot usage correlated with increased loneliness, emotional dependence, and problematic use - Personal conversation topics increased loneliness while non-personal topics increased dependence among heavy users - Participants with stronger emotional attachment tendencies experienced greater loneliness and emotional dependence

2. Anthropomorphism Measurement - Individual differences in anthropomorphism significantly predict social connection to AI companions, with 58-72% of participants scoring above thresholds for increased connection - Validated scales exist including the 6-item Perceived Anthropomorphism of Personal Intelligent Agents Scale and 8-item Robot's Perceived Empathy scale

3. Attachment & Dependency Patterns - Research using attachment theory scales found 52% of participants reported proximity seeking to AI, 77% used AI as safe haven, and 75% as secure base - Study of 54 Chinese adults found attachment anxiety positively predicts problematic CAI use, mediated by emotional attachment and moderated by anthropomorphic tendency

4. Usage Frequency & Temporal Factors - Study of 3,270 German adults found individuals using AI tools at least once weekly for personal conversation showed markedly poorer social disconnectedness outcomes - Social anxiety, loneliness, and rumination contribute to problematic CAI use through serial mediation

Recommended Improvements to Formula

1. Add Conversation Content Classification

$$I_{\text{message}} = \dots + (\text{Contenu}_{\text{personnel}})^{\lambda_{13}} + (\text{Contenu}_{\text{émotionnel}})^{\lambda_{14}}$$

Justification: Research shows personal topics increase loneliness while non-personal topics increase dependence differently

2. Include Attachment Style Pre-condition

$$I_{\text{global}} = \text{Style}_{\text{attachement}}^{\lambda_{15}} \times (\sum I_{\text{conversation}})^{\lambda_{12}}$$

Justification: Attachment anxiety significantly moderates the relationship between AI interaction and problematic use

3. Add Mind Perception/Theory of Mind

$$I_{\text{message}} = \dots + (\text{Perception}_{\text{esprit}})^{\lambda_{16}}$$

Justification: Mind perception intensifies the effect of social anxiety on problematic CAI use and buffers the association between rumination and dependency

4. Include Interaction Modality Weight

$$I_{\text{message}} = \text{Modalité}^{\lambda_{17}} \times [(\text{Voc}_{\text{relationnel}} \times \text{Implication}_{\text{IA}})^{\lambda_1} + \dots]$$

Justification: Voice-based interactions show different psychosocial effects compared to text, especially at high usage levels

5. Add Baseline Loneliness/Depression Factor

$$I_{\text{global}} = (\text{Solitude}_{\text{initiale}})^{\lambda_{18}} + \text{Style}_{\text{attachement}}^{\lambda_{15}} \times (\sum I_{\text{conversation}})^{\lambda_{12}}$$

Justification: The lonelier people are initially, the more problematic their usage becomes - loneliness is both cause and effect

6. Incorporate Self-Disclosure Intensity

$$I_{\text{message}} = \dots + (\text{Intensité}_{\text{divulgateion}})^{\lambda_{19}}$$

Justification: Users frequently self-disclose vulnerable thoughts to chatbots, and vulnerability exposure without emotional risk may reduce capacity for deep human connection

7. Add Discontinuity Stress Factor

$$I_{\text{temporel}} = \dots + (\text{Changement}_{\text{personnalité_IA}})^{\lambda_{20}}$$

Justification: Heavy users prefer consistency and become frustrated when chatbots forget past selves or change personality

Enhanced Formula Structure

Complete Revised Formula

$$I_{\text{message}} = \text{Modalité}^{\lambda_{17}} \times \left[(\text{Voc}_{\text{relationnel}} \times \text{Implication}_{\text{IA}})^{\lambda_1} + (\text{demande}_{\text{opinion}})^{\lambda_2} + (\text{mémorisation}_{\text{distinction}})^{\lambda_3} + \dots \right]$$

$$I_{\text{temporel}} = \text{Durée}_{\text{session}}^{\lambda_8} + \text{heure}_{\text{tardive}}^{\lambda_9} + \text{espacement}_{\text{précédent_dernière_conv}}^{\lambda_{10}} + (\text{Changement}_{\text{personnalité_IA}})^{\lambda_{20}}$$

$$I_{\text{conversation}} = \sum I_{\text{message}} + (\text{Nombre}_{\text{messages_fermeture}})^{\lambda_{11}} + I_{\text{temporel}}$$

$$I_{\text{global}} = \left[(\text{Solitude}_{\text{initiale}})^{\lambda_{18}} + \text{Style}_{\text{attachement}}^{\lambda_{15}} \right] \times (\sum I_{\text{conversation}})^{\lambda_{12}}$$